The University of Manchester

Individual 3rd Year Project (EEEN30330) Handbook

Department of Electrical and Electronic Engineering

School of Engineering

Faculty of Science and Engineering

2025 - 2026

Unit Organiser

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1. Key Dates and Deliverables

Semester 1			
Week	**W/C**	**Key Events**	**Deliverables**
0	22 Sep 2025	Project allocation	
1	29 Sep 2025	Project Meeting & Lecture 1: General Intro.	
2	06 Oct 2025	Project Meeting & Lecture 2: Intro. to LaTex and mathematical writing Intro. to CPD and Project/Time Management	
3	13 Oct 2025	Project Meeting	Submit **Preliminary Project Proposal** (project outline and project plan) and **H&S Risk Assessment** on CANVAS by 2pm on 17th Oct 2025
4	20 Oct 2025	Project Meeting including: - Review and approve of Preliminary Project Proposal - Review and approve of H&S Risk Assessment Supervisors should access Canvas and enter '1' to confirm their agreement.	
5	27 Oct 2025	Progress Review Meeting 1 Individual performance and technical progress	
6	03 Nov 2025	Project Meeting	
7	10 Nov 2025	Project Meeting & Lecture 3: Writing of Draft Introduction and Draft Methodology	
8	17 Nov 2025	Project Meeting	
9	24 Nov 2025	Project Meeting	
10	01 Dec 2025	Progress Review Meeting 2 Individual performance and technical progress	
11	08 Dec 2025	Project Meeting	
12	15 Dec 2025	Project Meeting	Submit **Draft Introduction and Project Methodology**. Completing **Academic Malpractice Awareness** training on CANVAS by 2pm on 19th Dec 2025

Semester 2			
Week	**W/C**	**Key Events**	**Deliverables**
1	02 Feb 2026	Project Meeting including: - Academic Malpractice Awareness supervisors record the result on Canvas - Review of Draft Introduction and Methodology & Lecture 4: Writing of the Draft Report/ Writing of the Final Report	
2	09 Feb 2026	Project Meeting	
3	16 Feb 2026	Progress Review Meeting 3 Individual performance and technical progress	
4	23 Feb 2026	Project Meeting	
5	02 Mar 2026	Project Meeting	
6	09 Mar 2026	Project Meeting	Submit **Draft of the Final Report** (partly populated) on CANVAS by 2pm on 13th Mar 2026
7	16 Mar 2026	Progress Review Meeting 4 Individual performance and technical progress	
8	23 Mar 2026	Project Meeting including - Review of the Draft of the Final Report (partly populated) - Review and agreed table of content	
9	20 April 2026	Project Meeting & Lecture 5: Prep. for Demonstration/Presentation	
10	27 April 2026	Project Meeting	
			Submit **Final Report** on CANVAS by 2pm on 1st May 2026
11	5 May 2026	Final Demonstration/Presentation	
12	11 May 2026		

2. Overview

The Individual 3rd Year Project is a mandatory, 30-credit component of the undergraduate programme and constitutes 25% of the overall assessment for Year 3. The project provides students with the opportunity to undertake an in-depth individual investigation within an area relevant to the disciplines taught within the Department of Electrical and Electronic Engineering. Projects may span a wide range of topics and are intended to reflect both academic and applied aspects of electrical, electronic and mechatronic engineering and related fields.

Academic staff propose a set of project topics aligned with their expertise. These topics are made available to students prior to project allocation. Students are invited to express their preferences by selecting up to six topics. Project allocation seeks, as far as reasonably practicable, to take student preferences into account while also ensuring an equitable distribution of supervisory workload.

Following allocation, each student is assigned a Project Supervisor. The precise project scope, aims and objectives are defined and refined through discussion between the student and the supervisor. Students allocated to a project that was one of their stated choices will not normally be permitted to reallocate to a different supervisor.

Projects are undertaken across two semesters. During the early stages, students receive closer guidance from their supervisor alongside undertaking initial background research. As the project progresses, students are expected to work with increasing independence, addressing the specific technical challenges and objectives defined for their individual project.

Student engagement and technical progress are continuously monitored by the Project Supervisor through regular weekly meetings. These meetings provide opportunities for formative feedback on progress and on preliminary written deliverables that are expected to form the basis of the Final Project Report. In addition, a subset of these meetings are designated as Progress Review Meetings, during which student engagement and progress are assessed summatively and contribute to the overall project mark. By the end of the project, students are expected to demonstrate independent learning capability, effective problem-solving skills, and the ability to communicate technical outcomes clearly and professionally.

The assessment of the Individual 3rd Year Project is based on the following components, each expressed as a percentage of the overall project mark:

- **Project Engagement and Progress:** 15%
- **Final Report:** 75%
- **Final Demonstration/Presentation:** 10%

Marking of *Project Engagement and Progress* is undertaken by the Project Supervisor during the Progress Review Meetings.

The *Final Report* is assessed independently by the Project Supervisor and an Independent Marker, with marks subsequently reviewed and agreed through a moderation process.

The *Final Demonstration/Presentation* is assessed jointly by the Project Supervisor and an Independent Marker.

Students must achieve an overall mark of at least 40% in the Individual 3rd Year Project in order to be awarded an Honours degree. Students who do not achieve this threshold will be considered for the award of an Ordinary degree, in accordance with University regulations.

Late submission of the Final Report will be penalised in accordance with the University's guidance on late submission, unless mitigating circumstances or DASS extension apply.

Mitigating circumstances must be brought to the attention of the Student Welfare Officer (soe.wellbeing@manchester.ac.uk) before the relevant deadline. Further information, including the Mitigating Circumstances form, is available in the Undergraduate Programme Handbook and on the EEEN30000 Virtual Common Room on Canvas.

The successful completion of the Individual 3rd Year Project is a shared responsibility between the student and the Project Supervisor. Students are expected to take primary responsibility for the planning, execution, and timely completion of their project, and to engage proactively with their supervisor throughout the project period. Project Supervisors are responsible for providing appropriate academic guidance, monitoring progress, and offering formative feedback in line with the expectations of an individual project.

The role of the Project Organiser is to provide overall coordination and oversight of the Individual 3rd Year Project process and to ensure consistency and fairness in its delivery across the student cohort. The Project Organiser does not replace the role of the Project Supervisor and is not involved in the day-to-day supervision of individual projects.

Any issues should, in the first instance, be discussed with the Project Supervisor. If a matter cannot be resolved at this level, or if the Project Supervisor is not an appropriate point of contact, students should raise the issue with the Project Organiser (sareh.malekpour@manchester.ac.uk) as soon as possible.

3. Meetings with the Project Supervisor

3.1. Regular Weekly Project Meetings

Students are expected to engage in **regular weekly project meetings** with their Project Supervisor throughout the duration of the project. The precise scheduling and format of meetings should be agreed between the student and the Project Supervisor, considering the nature of the project and practical constraints.

If a student is unable to attend a scheduled meeting due to exceptional circumstances, the Project Supervisor should be informed at the earliest opportunity so that the meeting can be rearranged. Additional one-to-one meetings may be arranged by mutual agreement between the student and the Project Supervisor where appropriate.

With the exception of the initial meeting, project meetings must be individual and held in person.

During project meetings, students are expected to be proactive and to take primary responsibility for managing their project activities. Students are responsible for keeping a record of meetings, agreed actions, and project progress. Students are also expected to record activities related to self-directed learning and skills development informed by technical literature, relevant codes of practice, and industry standards. Students are, therefore, strongly encouraged to maintain a project logbook to document project activities, agreed actions, and self-learning. The logbook is intended as a developmental tool and does not itself constitute a credit-bearing assessment component.

3.2. Progress Review Meetings (PRMs)

Four of the regular project meetings (two in Semester 1 and two in Semester 2) are designated as formal Progress Review Meetings (PRMs). These meetings form the basis for assessment of Project Engagement and Progress.

At each Progress Review Meeting, the Project Supervisor assesses the student's engagement and technical progress and provides a mark accompanied by written feedback. Feedback is normally provided before the next scheduled project meeting.

The assessment focuses on:

- the student's level of engagement and professionalism,
- technical progress relative to the agreed project scope,
- the student's ability to plan and execute self-directed learning and skills development required for the project.

This last aspect of planning and executing self-directed learning aligns with Continuing Professional Development (CPD) principles, which are introduced further in Lecture 2 delivered on Monday 6th of October 2025 at 2pm-4pm.

Intermediate deliverables (such as Draft Introduction, Draft Project Methodology and draft sections of the Final Report) may be discussed during Progress Review Meetings. However, the Project Engagement and Progress mark is based on engagement and progress, not on the quality of these intermediate deliverables. Feedback on these deliverables is formative only and does not contribute to credit-bearing assessment.

Each Progress Review Meeting is marked on a scale of 0–10 (integer values). Feedback and marks are recorded in CANVAS. The final Project Engagement and Progress mark is calculated by aggregating the marks from all four PRMs and contributes 15% to the overall project mark.

Examples:

- A student awarded 8/10 in each of the four PRMs achieves a contribution of 12% (out of 15%) towards the total project mark.
- A student awarded 4/10, 6/10, 6/10, and 8/10 across the four PRMs achieves a contribution of 9% (out of 15%) towards the total project mark.

The marking criteria for Progress Review Meetings are provided in the table on the following page.

Mark range	Engagement and Technical Progress
0	No attendance; No engagement in project work; No evidence of self-learning and skill development;
1-2	Low level of attendance and/or consistently late in the meetings; Level of effort and interest in the project is substantially below standard; Little contributions of ideas; Little evidence of engagement in project work; Inadequate progress; Little evidence of self-learning and skill development;
3-4	Some attendance and/or consistently late in the meetings; Limited or adequate levels of effort; Some contributions of ideas; Evidence of engagement in project work is marginal; Limited performance that is barely acceptable; Some but limited evidence of self-learning and skill development;
5-6	Good attendance or attendance OK but occasionally late; Good levels of effort; Good contributions of ideas; Good performance and show evidence of exploiting potential opportunities; Substantial evidence of both planning and executing self-learning and skill development;
7-8	Exemplary attendance; Excellent level of effort; Fully prepared and proactively engaged in the technical discussion; Fully engaged in the project work; Significant independent contribution to development of project; Capable of independent progress to a large extent; Proactive engagement in planning and highly effective execution of self-learning and skill development;
9-10	Exemplary attendance; Outstanding project management and technical progress that substantially exceeds expectations from the Project Supervisor; Exceptional ability to plan and execute self-learning and skill development;

4. Academic Malpractice Awareness

The Individual 3rd Year Project is an assessment of both technical competence and professional academic practice. All submitted work must comply with the University's regulations on academic integrity, including appropriate referencing and avoidance of plagiarism.

To support students in meeting these expectations, the University requires completion of **Academic Malpractice Awareness** training prior to the submission of the Final Report. Students are strongly advised to complete this training as early as possible and not to leave it until close to the Final Report submission deadline.

To satisfy the Academic Malpractice Awareness requirement, students must:

a) Complete the "*Short Course on Academic Malpractice*" on CANVAS.

b) Pass the '*Driving Test*' at the end of the "*Short Course on Academic Malpractice*".

c) Complete the *Self Declaration Form* that appears after passing the '*Driving Test*'.

d) Complete the *Tutorial Questions* and provide them to the Project Supervisor.

Once the Project Supervisor is satisfied that these tasks have been successfully completed, they will enter a '1' in the "*Tutorial Questions*" column against student name in the Grade Centre on CANVAS.

Students should note that **submission of the Final Report will not be permitted** until the Academic Malpractice Awareness requirements have been completed. Failure to complete these requirements in advance of the Final Report deadline may therefore result in **late submission penalties** being applied.

Students must ensure that all submitted work is their **own**, except where clearly indicated and appropriately referenced. This includes proper paraphrasing of ideas and concepts drawn from other sources.

Direct quotations from books, journal articles, websites, or other sources must be placed within quotation marks and accompanied by an appropriate citation.

All cited material must be fully referenced in the References section of the Final Report.

Plagiarism, whether deliberate or accidental, constitutes a serious breach of academic integrity and may result in disciplinary action in accordance with University regulations. Full details of the University's policy on plagiarism are available at documents.manchester.ac.uk/DocuInfo.aspx?DocID=2870..

POWERFUL SOFTWARE TOOLS AVAILABLE WITHIN CANVAS (TURNITIN) WILL BE USED TO INSPECT EVERY REPORT SUBMITTED FOR PLAGIARISM. UNIVERSITY REGULATIONS DO NOT ALLOW STUDENTS ACCESS TO THE RESULTS OF TURNITIN.

5. The Preliminary Project Proposal and Risk Assessment

Each student must prepare a 2-page Project Outline that describes the background, motivation for the project, overall aim and the set of associated objectives (milestones) of the project. The purpose of the Project Outline is to clearly define the initial scope of the project and identify what needs to be achieved in order to successfully complete the project. It also allows supervisor to provide feedback on the student's understanding of the project scope.

It is recognised that **the scope may change during the course of the project** as a result of ongoing discussions between the student and the Project Supervisor.

The Preliminary Project Proposal should also include a **(provisional) Project Plan**. This plan should set out the intended timeline for the project, typically in the form of a GANTT chart, and include a brief description of each major task (nominally few sentences per task). The plan is also expected to identify the key risks to the successful completion of the project, together with strategies to mitigate or manage those issues that cannot be prevented. It is expected that the Project Plan will be reviewed and updated as the project progresses.

In addition, each student is required to complete a **Health & Safety Risk Assessment**, regardless of whether project involves handling of hazardous equipment/materials, or is primarily computer-based. The Risk Assessment identifies potential health and safety hazards associated with the project and the measures in place to eliminate or reduce those risks. Students should refer to the online health and safety induction course, which is delivered at the beginning of Semester 1, when preparing the Risk Assessment.

- **RISK ASSESSMENT MUST BE COMPLETED AND APPROVED BY THE PROJECT SUPERVISOR BEFORE ANY PRACTICAL WORK CAN TAKE PLACE.**

The Preliminary Project Proposal (including the Project Outline and Project Plan) and the Health & Safety Risk Assessment should be submitted on CANVAS **before 2pm on Friday 17th of October 2025** so that they can be discussed during the Progress Review Meeting 1 which will take place in Semester 1 Week 5 (week commencing on the 27th of October 2025).

There is no formal late-submission penalty associated with these deliverables. However, failure to submit them by the stated deadline is likely to have a negative impact on the *Project Engagement and Progress* mark awarded at Progress Review Meeting 1 (PRM1), as it may indicate a lack of engagement with the project.

6. Draft Introduction

Students are expected to prepare a Draft Introduction document, structured in the same way as the introductory chapter of the Final Report. The Draft Introduction should be no more than 1,000 words.

The purpose of the Draft Introduction is to provide an opportunity for formative feedback on the clarity of the project context, motivation, aims, and overall framing of the work at an early stage of the project. The Draft Introduction does not contribute to the final project mark and is not a summative assessment component.

Draft Introduction should be submitted on CANVAS **before 2pm on Friday 19th of December 2025** so that it can be discussed during the Project Meeting which will take place in Semester 2 Week 1 (week commencing on the 2nd of February 2026). There is no formal penalty for late or no submission of the Draft Introduction, but it will affect the quality of feedback provided by the Project Supervisor to the

student.

Further guidance on the expected structure and content of the Draft Introduction will be provided during Lecture 3, timetabled on Thursday 13th of November 2025, 3pm–4pm.

7. Draft Project Methodology

The Project Methodology should address both the **"how"** and the **"why"** of the project execution. This includes explaining how particular elements of the work are undertaken (for example, theoretical development, system design, modelling, simulation, implementation, or experimentation), together with a justification for the chosen approach. The rationale for the methodology should be linked to the project background and motivation.

Students are required to prepare a Draft Project Methodology for the purpose of obtaining formative feedback. This deliverable does not contribute to the final project mark and is not a summative assessment component.

The Draft Project Methodology should be submitted (together with the Draft Introduction) via CANVAS by **2pm on Friday 19th of December 2025**, in advance of the Project Meeting taking place in Semester 2, Week 1 (week commencing 2 February 2026), during which feedback from the Project Supervisor is expected to be discussed.

There is no formal penalty for late submission or non-submission of the Draft Project Methodology. However, failure to submit this material by the stated deadline is likely to limit the amount and usefulness of feedback that can be provided by the Project Supervisor at this stage of the project.

Further guidance on the expected structure and content of the Project Methodology will be provided during the lecture timetabled on Thursday 13th of November 2025, 3pm–4pm.

8. Draft of the Final Report

Similarly to Draft Introduction and Draft Project Methodology, students are expected to submit a draft of the Final Report for the purpose of obtaining formative feedback. This submission does not contribute to the final project mark and is not a summative assessment component.

The draft of the Final Report should be submitted on CANVAS **before 2pm on Friday 13th of March 2026.**

The draft report should follow the overall structure of the final report and should include, if available, most of the following sections:

- **Table of Contents**
- **Introduction**
- **Methodology**
- **Results**
- **Conclusion and Future Work**

The draft does not need to be complete. However, students are strongly encouraged to include the full set of intended section and sub-section headings, even where content is incomplete, to enable feedback on the overall structure and coherence of the report.

Written formative feedback will be provided by the Project Supervisor on the Table of Contents and on one substantial section or chapter of the student's choosing, amounting to no more than 10 pages in total. Students should indicate clearly which section they wish to receive detailed feedback on at the point of submission. Since feedback will already have been provided on the Draft Introduction (Semester 1) and the Draft Project Methodology (Semester 2), students are encouraged to select the Results chapter for this additional feedback opportunity before submitting the Final Report.

9. The Final Report

9.1. Submission Details

Students must submit an electronic copy (PDF format) of the Final Report via CANVAS by:

2:00 PM on Friday the 1st of May 2026 (Friday in Semester 2 Week 10)

Students are encouraged to allow themselves enough time before the deadline to avoid late-submission penalties, as they can lose up to 10 marks for delays of only a few minutes. Please note that the resubmission of the thesis *after the deadline*, regardless of the extent of the changes, will also incur a late-submission penalty.

No hardcopy submission is required.

FINAL REPORTS SUBMITTED AFTER THE DEADLINE WILL BE PENALISED IN ACCORDANCE WITH THE UNIVERSITY'S GUIDANCE ON LATE SUBMISSION
(http://www.regulations.manchester.ac.uk/academic/late-submission/).

Extensions to the Final Report deadline are granted **only** where a DASS extension is in place or where mitigating circumstances have been formally approved in advance in accordance with University procedures.

9.2. Format of the Final Report

The Final Report **MUST** be word-processed and presented in accordance with the following requirements:

- The main body of the report including text, diagrams **MUST NOT EXCEED 35 NUMBERED A4** pages. The following are excluded from the page limit: abstract, table of contents, lists of figures/tables, glossary, references, and appendices.
  Penalty for exceeding maximum size is a deduction of 5 marks from the Final Report mark.

- **Margin must be at least 2 cm** on all sides (left, right, top, and bottom).

- The main body text must use **1.5 spacing**. Single spacing may be used for quotations, footnotes, and references.

- Text **must be left aligned** (not fully justified).

- The font type and size for the report should be **Calibri 12** (or a close equivalent).

The overall structure and formatting of the Final Report should normally follow the **template provided on CANVAS**. Some flexibility is permitted, but any significant deviations from the template should be discussed with the Project Supervisor in advance.

Further guidance on writing the Final Report will be provided during the lecture scheduled for **Friday 6th of February 2026, 3pm–4pm**.

Use of Intermediate Deliverables: Material previously submitted as formative or intermediate deliverables (such as the Draft Introduction, Draft Methodology, or Draft Final Report) may be reused and further developed in the Final Report. This will **not** be regarded as self-plagiarism. Students are expected to revise and refine this material in light of feedback received, ensuring that the Final Report represents a coherent, polished, and complete account of the project.

9.3. Project Management in the Final Report

Project management forms an integral part of the Individual 3rd Year Project and provides evidence of students' ability to plan, monitor, and reflect on their work in a structured and professional manner. As in the 2nd Year Embedded Systems Project (ESP), students are expected to demonstrate that project management has been used actively to support the successful delivery of the project, rather than being treated as a purely administrative exercise.

Project management in the main body of the report

Students must include a **dedicated Project Management section** within the main body of the Final Report. This section should normally be **no more than one page** in length and should provide a concise reflection on how the project was planned and managed over time.

In this section, students are expected to:

- describe how the project plan was used to structure and monitor progress,
- reflect on any significant deviations from the original plan and explain why changes were necessary,
- discuss how risks were identified and managed during the project,
- comment on their approach to self-directed learning and skills development where relevant,
- demonstrate initiative and personal responsibility in managing the project.

This discussion should draw explicitly on the project management artefacts provided in the appendices.

Project management artefacts (appendices)

The following project management artefacts **must** be included in the appendices of the Final Report:

- **Project Plan and GANTT Chart (final, updated version)**
- **Risk Register**
- **Continuing Professional Development (CPD) Log**
- **Health & Safety Risk Assessment**

These artefacts provide supporting evidence for the project management discussion in the main report and are not assessed in isolation.

The Project Plan and GANTT Chart included in the Final Report should represent the **final, updated version** of the plan, reflecting how the project was actually executed. Students are not assessed on how closely the original plan was followed, but on how effectively the plan was used as a tool for:

- organising work,
- monitoring progress,
- adapting to unforeseen challenges, and
- communicating progress to stakeholders (e.g. the Project Supervisor).

Similarly, the Risk Register should demonstrate that risks were identified, reviewed, and managed throughout the project lifecycle, and the CPD Log should provide evidence of planned and executed self-learning aligned with the technical needs of the project.

9.4. Appendices

Appendices are intended to provide **supporting material** that underpins the work presented in the main body of the Final Report. While material included in appendices does **not contribute directly** to the Final Report mark, appendices play an important role in evidencing the quality, depth, and integrity of the work described in the report.

Material included in appendices is not assessed in isolation and does not attract marks in its own right. In particular:

- appendices should not be used to introduce new core results,
- key arguments, results, and conclusions must appear in the main body of the report,
- appendices must not be used to circumvent the page limit of the main report.

However, appendices may be consulted by markers to:

- verify and contextualise results presented in the main report,
- assess the appropriateness and robustness of methods used,
- confirm that professional and safety requirements have been addressed.

In this way, appendices support the assessment of the main report without forming a separate, credit-bearing component.

The following items **must** be included in the appendices of the Final Report (:

- **Preliminary Project Proposal**
- **Project Plan and Gantt Chart**
- **Risk Register**
- **Continuing Professional Development (CPD) Log**
- **Health & Safety Risk Assessment**
- **Link to the version-controlled repository** containing software developed as part of the project (where applicable). Further details regarding this item are provided in Section 9.5.

Please note that not all students are required to include a repository link, particularly those undertaking hardware-focused projects. Therefore, a repository link is required only for students involved in software development.

Please note that failure to include these required elements will result in the application of penalties.

Students may also include additional supporting material in the appendices where this helps to substantiate or clarify claims made in the main body of the report. This may include, for example:

- extended derivations or proofs,
- supplementary experimental or simulation results,
- additional figures or tables that support the main findings,
- detailed algorithm descriptions or parameter listings.

Such material should be clearly referenced from the main report where relevant.

9.5. Code and Software Artefacts

Where relevant, students are required to submit source code using a **version-controlled repository** (e.g. GitHub or GitLab) and to include a **link to the repository** in the appendix of the Final Report.

The repository must be clearly organised and accompanied by a brief description of its structure, explaining the purpose of the main directories and files required to run or evaluate the work.

Any imported, reused, or third-party code must be clearly identified and appropriately referenced. Students remain responsible for ensuring that all submitted work complies with academic integrity requirements.

Further guidance on the submission of code and software artefacts is provided in the Appendix.

10. Assessment of the Final Report

10.1. Marking Process

The Final Report is assessed by the Project Supervisor and an Independent Marker. Each marker assesses the Final Report **independently** and submits their marks and feedback without discussion prior to submission. Marks are then reviewed through a moderation process and final mark is agreed in accordance with university procedures. This approach is intended to minimise bias and to ensure that no marker influences the judgement of the other.

10.2. Marking Criteria and Weightings

Overview, Background and Conclusions (25%)

This component assesses how effectively the project is motivated, contextualised and concluded. Assessed elements are:

- Abstract,

- Background and motivation,
- Statement of aim and objectives,
- Literature review and/or consideration of existing solutions,
- Conclusions,
- Project Management (and associated artefacts).

Technical Achievement (65%)

This component assesses the technical depth, rigour, and quality of the work undertaken. Assessed elements are:

- Theoretical development,
- Design,
- Implementation,
- Testing & Analysis,
- Results and Discussion.

Not all projects will include all of the above elements (for example, many of the projects may not involve significant theoretical development). Marks are awarded based on the appropriateness, quality, and execution of the technical work undertaken within the agreed project scope.

Presentation and Content (10%)

This component assesses the clarity and professionalism of the written report. Assessed elements are:

- Grammar and spelling
- Clarity and legibility of figures, tables, and diagrams
- Consistency and quality of formatting
- Logical structure and organisation of the report
- Appropriate use of sections, appendices, and references

11. The Final Demonstration/Presentation

The Final Demonstration/Presentation takes place during **Week 11 or Week 12 of Semester 2** and provides students with the opportunity to present and demonstrate the outcomes of their Individual 3rd Year Project.

The Final Demonstration/Presentation is attended by the student, the Project Supervisor, and an Independent Marker. Individual time slots will be scheduled between **5th of May 2026 and 15th of May 2026**.

Each student is required to deliver a demonstration/presentation of no more than 15 minutes, followed by up to 15 minutes of questions, giving a total maximum duration of **30 minutes**. Students are expected to focus on their main technical achievements and, where appropriate, to use quantitative or scientific indicators to support their claims.

The Final Demonstration/Presentation is assessed by the Project Supervisor and the Independent Marker. Marks and feedback are agreed following the demonstration/presentation in accordance with Departmental procedures.

Students are assessed on their ability to:

- provide a clear overview of the project and its objectives,
- demonstrate and explain the main technical achievements of the project,
- respond effectively to questions about their work.

Further guidance on preparing for the Final Demonstration/Presentation will be provided during the lecture scheduled for Friday 24th of April 2026, 3pm–4pm.

Final Demonstrations/Presentations are normally conducted **in person**, with the student, Project Supervisor, and Independent Marker all attending.

If a student wishes to use a laboratory or other specific location (for example, to demonstrate hardware or experimental work), then this must be discussed with the Project Supervisor well in advance of the scheduled demonstration. Appropriate arrangements must be made jointly to ensure that the demonstration can take place safely and effectively in the chosen venue.

Attendance at the Final Demonstration/Presentation is **mandatory**. If a student is unable to attend their scheduled demonstration due to exceptional circumstances, they must notify the Project Supervisor as soon as possible to discuss re-arrangement.

Failure to attend the Final Demonstration/Presentation without an approved reason will result in a mark of **zero** for this assessment component.

Please note that DASS deadline extensions apply to written coursework and therefore **do not apply** to the Final Demonstration/Presentation, which is a live, timetabled assessment. Any inability to attend a scheduled demonstration due to exceptional circumstances must be addressed through the University's mitigating circumstances process.

12. Marking Norms

The marking norms presented in Sections 12.1 and 12.2 provide **indicative guidance** on the standards expected at different performance levels for the final project deliverables, namely the Final Report and the Final Demonstration/Presentation.

These norms are used by examiners to support consistent and fair assessment across projects. They are not intended as a checklist, and individual projects may demonstrate strengths and weaknesses in different ways.

Students are encouraged to use these marking norms as general guidance when preparing their final deliverables, alongside the marking criteria and weighting described in the previous sections.

12.1. Marking Norms for the Final Report

Mark	Category	Y3 Individual Project **Final Report**
90-100	Exceptional 1st Class	*Adjectives to be used to describe the work/performance: exceptional, faultless* Students achieving marks in this grade range should have satisfied all conditions in the previous grade range and in addition the following conditions must be met: -Exceptional work of the highest quality in all criteria of assessment displaying significant originality and/or deep insight. Likely to be of publishable quality. -Examiners must be convinced that this is new material that is on the boundaries of existing knowledge in the field.
80-89	Outstanding 1st Class	*Adjectives to be used to describe the work/performance: outstanding, remarkable, exceeding* Students achieving marks in this grade range should have satisfied all conditions in the previous grade range and in addition the following conditions must be met: -Outstanding work of the highest quality demonstrating comprehensive knowledge, excellent critical analysis and/or originality, high level of accuracy, relevance, presentation, and appropriate skills. - Work marked in this range will demonstrate intellectual originality and imagination, and include innovative development and/or analysis.
70-79	1st Class	*Adjectives to be used to describe the work/performance: excellent, comprehensive, rigorous* -Excellent work of high quality, demonstrating extensive knowledge, very good critical analysis, a high level of accuracy, relevance, presentation, and appropriate skills. -Excellent technical achievement, demonstrating skilful application of a wide range of appropriate engineering tools and techniques. -The student should also be able to situate their work within the wider context, showing an awareness of current state-of-the-art and an ability to relate their results to the work of others, making full and appropriate use of references. -The report will be well-presented, written in a clear and concise manner, with a clear and logical structure.
60-69	Upper 2nd Class (2i)	*Adjectives to be used to describe the work/performance: very good, effective* -High quality work, demonstrating very good knowledge and understanding. -Very good level of technical achievement, with proficient application of relevant engineering skills. -Good critical analysis, accuracy, relevance, and presentation. -The work will be competently or well presented, making effective use of references to show the backdrop of the work. -The work will be clearly and effectively written, with technically correct descriptions.
50-59	Lower 2nd Class (2ii)	*Adjectives to be used to describe the work/performance: good, satisfactory, competent* - Competent and generally accurate work, demonstrating some relevant knowledge, sound understanding and good technical

		achievement, though undeveloped with limited critical reasoning. - There should be some attempt to situate the work within a wider context, but the presented background may be relatively narrow. - Research, referencing and prose styles may be haphazard, and arguments may be one-sided. - The work should be competently presented but may not entirely follow the guidelines set.
40-49	3rd Class	Adjectives to be used to describe the work/performance: acceptable, adequate, sufficient NOTE: Students that receive an overall project mark in this category would not be allowed to progress to Year 4 of the MEng course. -Adequate work, demonstrating some relevant knowledge and satisfactory technical achievement, but with limited understanding and limited critical reasoning. -There is no attempt to situate the work within a wider context. -Research, referencing, and prose styles are very limited. -The work is poorly presented and does not follow the guidelines set.
30-39	Compensatable Fail	*Adjectives to be used to describe the work/performance: <u>inadequate, insufficient</u>* -Work that falls below the standard required for BEng, but which shows some evidence of basic understanding and/or achievement. **NOTE:** If the **overall mark for the project** is in this category the student would not qualify for the award of a Beng/MEng Honours degree, but may still be able to graduate with an Ordinary degree.
0-29	Fail	*Adjectives to be used to describe the work/performance: <u>inadequate, insufficient</u>* -Work that substantially falls below the standard required for BEng, and is also seriously deficient in quality and quantity. **NOTE:** If the **overall mark for the project** is in this category the student would not qualify for the award of a Beng/MEng Honours degree, but may still be able to graduate with an Ordinary degree.

12.2. Marking Norms for the Final Demonstration/Presentation

Mark	Category	Y3 Individual Project **Final Demonstration/Presentation**
90-100	Exceptional 1st Class	Students achieving marks in this grade range should have satisfied all conditions in the previous grade range and in addition the following conditions must be met: -The work is presented/demonstrated throughout in a manner which is entirely suited to the subject-matter and allows for a powerful engagement with the relevant audience. -Oral expression and interaction with the audience show considerable flair. -The handling of follow-up questions corresponds to professional standards.
80-89	Outstanding 1st Class	Students achieving marks in this grade range should have satisfied all conditions in the previous grade range and in addition the following conditions must be met: -Answers to follow-up questions are used to develop lines of argument further and contribute substantially to discussion of the topic at hand.
70-79	1st Class	-The work is presented/demonstrated throughout in a manner which is very well suited to the subject-matter and allows for an engagement with the relevant audience. -The student keeps an appropriate balance between analysis and example and is able to reinforce key points without labouring them. -Answers to follow-up questions reveal a good range and depth of knowledge beyond that covered in the presentation and show confidence in discussion of abstract ideas.
60-69	Upper 2nd Class (2i)	-The work is well presented/demonstrated, in a way which allows for some engagement with the relevant audience. -The student highlights key points appropriately and keeps to the allotted time. -Able to draw on appropriate knowledge to answer follow-up questions; answers are presented clearly.
50-59	Lower 2nd Class (2ii)	-The work is presented/demonstrated in a competent manner, but in a way that provides a limited engagement with the relevant audience. -The student may demonstrate poor time-keeping. -Tendency towards narrative/description with little, if any analysis, particularly at the lower end of the range, where work is likely to contain material which is irrelevant. -Some hesitation in answering follow-up questions and/or gives incomplete or partly irrelevant answers, but nevertheless able to offer some appropriate information or views.
40-49	3rd Class	-The presentation/demonstration is largely unstructured, and some points are irrelevant to the topic. -The student may read from a script, making little attempt to address their ideas to the group; they may express themselves poorly or too colloquially. -Brief and sketchy answers to follow-up questions, while offering some basic information/opinions, reveal gaps in knowledge and understanding.
30-39	Compensatable Fail	-The presentation/demonstration is unstructured, and most points are irrelevant to the topic. -The student may read from a script, making no attempt to address their ideas to the examiners

		-Brief and sketchy answers to follow-up questions, revealing considerable gaps in knowledge and understanding.
0-29	Fail	-Only the barest attempt, or no attempt to address the topic and explain it to an audience. -Unable to answer follow-up questions.

13. Mitigating Circumstances

If a student is unable to submit intermediate formative project deliverables (for example, materials associated with Progress Review Meetings) by the stated deadlines, they should inform their Project Supervisor as soon as possible. In such cases, the Project Supervisor may allow limited flexibility, where appropriate.

There is no flexibility associated with the deadlines for submission of the Final Report or delivery of the Final Demonstration/Presentation, except where:

- a DASS deadline extension applies to the Final Report, or
- mitigating circumstances or extension request have been formally approved in accordance with University procedures.

DASS deadline extensions apply to written coursework submissions only and do not apply to live, timetabled assessments such as the Final Demonstration/Presentation.

Where mitigating circumstances affect a student's ability to meet the Final Report or Final Demonstration/Presentation deadlines, a **Mitigating Circumstances Form** or **Extension Request** must be submitted. Students are strongly advised to seek guidance as early as possible from the Student Welfare Officer (soe.wellbeing@manchester.ac.uk).

PLEASE NOTE THAT THE LOSS OF DATA OR TEXT DUE TO HARD DISK/COMPUTER FAILURE IS NOT RECOGNISED AS A VALID MITIGATING CIRCUMSTANCE. YOU SHOULD REGULARLY BACKUP YOUR DATA ON AT LEAST ONE OTHER DEVICE. SERVICES SUCH AS CLOUD STORAGE MAKE THIS EASY. IT IS NOT UNCOMMON FOR COMPANIES TO SACK EMPLOYEES WHO FAIL TO MAKE ADEQUATE BACK-UPS – SO GET INTO THE HABIT NOW.

14. Appendix - Evidencing Software in the Final Report

Where a project involves the development or modification of software, students are required to provide appropriate evidence of this work as part of the Final Report.

14.1. How software should be evidenced

Source code should not normally be embedded directly in the Final Report or appendices, as this can be difficult to inspect and reuse when converted to document format. Instead, students are expected to use a **version-controlled code repository** (for example, GitHub, GitLab, or Bitbucket) and to include a link to this repository in the appendices of the Final Report.

The repository must be **publicly accessible** at the point of submission.

Rather than including full source code listings in the report, students should include:

- a **link to the repository**, and
- a clear **description of the software**, provided via a **README file**.

This approach supports inspection, reuse, and reproducibility of the work.

14.2. What is meant by a code repository

A **repository (repo)** is a structured project space that contains all files associated with a software project, typically organised within a single directory. Repositories are commonly managed using **version control**, which allows changes to be tracked over time.

Services such as **GitHub**, **GitLab**, and **Bitbucket** provide online hosting for repositories, enabling:

- secure backup of code,
- version tracking,
- access from multiple machines,
- sharing of code with others.

Students are not expected to be experts in version control but are expected to use a repository in a clear, organised, and professional manner.

14.3. Repository requirements

For each software repository referenced in the Final Report, students must provide:

- A **direct link** to the publicly accessible repository
- A **README file**, included in the repository and referenced in the appendix

If multiple repositories are used, each must be clearly identified and explained.

14.4. README file: required content

Each repository must contain a README file. The README should be written so that a technically competent reader with no prior knowledge of the project can understand and run the software.

As a minimum, the README should include:

1. **Title** – Name of the project or software component.
2. **Introduction** – A brief description (1–2 paragraphs) of what the software does and its role within the overall project.
3. **Contextual overview** – A diagram, image, or short explanation that situates the software within the project (e.g. system architecture, data flow, or control structure).
4. **Installation instructions** – Step-by-step instructions describing:
    - required software or libraries,
    - dependencies,
    - environment setup.
5. **How to run the software** – Clear instructions or a simple example demonstrating how the software is executed.
6. **Technical details** – Any relevant technical information required to understand the implementation, such as:
    - algorithms,
    - equations,
    - design assumptions.
7. **Known issues and future improvements** – A brief list of limitations, bugs, or potential extensions.

The README should be clearly formatted and legible. Minor formatting differences between the repository and the appendix are acceptable.

14.5. Academic integrity and third-party code

Any **imported, reused, or third-party code** must be:

- clearly identified,
- appropriately referenced,
- used in accordance with relevant licences.

Students remain responsible for ensuring that all submitted software complies with University regulations on academic integrity.